

1. List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

Ans. Device that can work on the **SMPS**

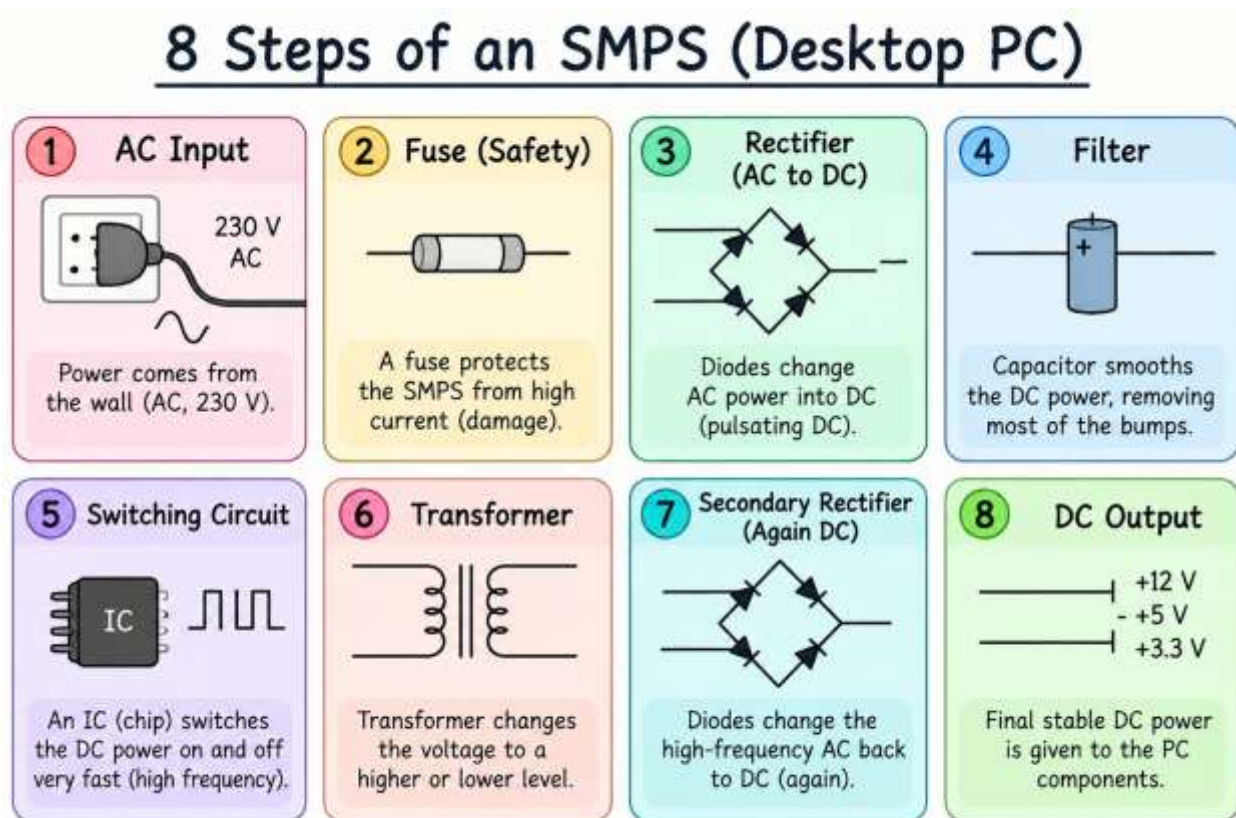
Device	Why it uses SMPS
Laptop charger	It can convert the AC power to DC power to needs the laptop
Desktop computer (Cabinet)	They can give the stable power to the motherboard, processor and other parts.
TV	It can convert the electricity into the right voltage for to work safely.

Device work on the UPS

Device	Why it uses UPS
Desktop computer	It can help to running the computer for some time when lights turn off.
Wi-Fi router	It can give the internet connection for sometimes when lights get turn off.

2. Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections. You can hand-draw and upload a photo, or use any free diagram tool like draw.

Ans.



3. Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

Ans.



4. Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

Ans. Steps to connect the SMPS.

1. Switch off the pc and remove the power cable.
2. In Motherboard connect the 24 pin ATX cable to the motherboard.
3. Connect the 4/8 pin CPU cable near the processor.
4. Graphic card Connect the 6/8 pin PCIe cable to the GPU
5. SSD connect to the SATA power cable to the storage drive.
6. Check the all cables and make sure the all connectors are properly plugged in.
7. Than plug in the power cable and start the computer to check the everything works.

